



Dr. Vishwanath Karad
MIT WORLD PEACE
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TECHNOLOGY, RESEARCH, SOCIAL INNOVATION & PARTNERSHIPS

Dr. Vishwanath Karad MIT - World Peace University
Faculty of Engineering and Technology
School of Computer Science and Engineering
Department of Computer Science and Applications
National Academic Immersion Program (NAIP)
National Institute of Technology, Goa
26 June To 7 July 2026

Submitted by :- Vedant Dilip Khedkar
PRN :- 1272240644
TYBCA-B

Day 1

Session 1

Field	Details
Date	6/27/2026
Day wise Session (Morning and afternoon)	Session 1 (Forenoon, 9.30 AM - 11.00 AM)
Name of the Resource Person	Person A
Topic Covered	Introduction to Deep Learning & AI Evolution
Brief Description	An introductory overview of deep learning principles and the historical timeline and evolution of artificial intelligence technologies.

Learning/Take Away	Gained a foundational understanding of AI's progression over time and the core concepts that define deep learning architectures.
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Session 2

Field	Details
Date	6/27/2026
Day wise Session (Morning and afternoon)	Session 2 (Forenoon, 11.30 AM - 1.00 PM)
Name of the Resource Person	Person A
Topic Covered	Applications of Generative AI
Brief Description	A discussion focused on the practical, real-world use cases and applications of Generative AI across various industries.
Learning/Take Away	Learned how generative models are deployed in practical scenarios and their impact on current technological solutions.

Session 3

Field	Details
Date	6/27/2026
Day wise Session (Morning and afternoon)	Session 3 (Afternoon, 2.00 PM - 3.30 PM)

Name of the Resource Person	Person A
Topic Covered	Setting up Python environment, and basic python programs
Brief Description	A practical session focused on installing and configuring the Python development environment, followed by writing fundamental Python scripts.
Learning/Take Away	Acquired hands-on experience in initializing a programming workspace and successfully executing basic Python code.

Day 2

Session 1

Field	Details
Date	6/28/2026
Day wise Session (Morning and afternoon)	Session 1 (Forenoon, 9.30 AM - 11.00 AM)
Name of the Resource Person	Person A
Topic Covered	Neural Networks (MLP, Perceptron)
Brief Description	An introduction to the fundamental concepts of Neural Networks, focusing on Multi-Layer Perceptrons (MLP) and basic Perceptron models.
Learning/Take Away	Understood the basic architecture of neural networks and how perceptrons function as the fundamental building blocks of these systems.

Session 2

Field	Details
Date	6/28/2026
Day wise Session (Morning and afternoon)	Session 2 (Forenoon, 11.30 AM - 1.00 PM)
Name of the Resource Person	Person A
Topic Covered	Activation & Loss Functions
Brief Description	A detailed look into various activation functions that introduce non-linearity, alongside loss functions used to measure and evaluate model performance.
Learning/Take Away	Gained the ability to identify and apply appropriate activation and loss functions depending on the specific requirements of a neural network task.

Session 3

Field	Details
Date	6/28/2026

Day wise Session (Morning and afternoon)	Session 3 (Afternoon, 2.00 PM - 3.30 PM)
Name of the Resource Person	Person A
Topic Covered	Gradient Decent & Backpropagation
Brief Description	An exploration of the core mechanisms of network training, specifically focusing on the gradient descent optimization algorithm and the backpropagation process.
Learning/Take Away	Comprehended how neural networks learn by iteratively updating their weights and biases to minimize errors during the training phase.

Session 4

Field	Details
Date	6/28/2026
Day wise Session (Morning and afternoon)	Session 4 (Afternoon, 4.00 PM - 5.30 PM)

Name of the Resource Person	Person A
Topic Covered	Hands-on: Build ANN
Brief Description	A practical, hands-on session dedicated to coding, constructing, and implementing an Artificial Neural Network (ANN).
Learning/Take Away	Successfully built and ran a basic ANN, translating theoretical concepts of nodes and layers into a functional programming project.

Day 3

Session 1

Field	Details
Date	6/29/2026
Day wise Session (Morning and afternoon)	Session 1 (Forenoon, 9.30 AM - 11.00 AM)
Name of the Resource Person	Person A
Topic Covered	Deep Neural Networks (DNN)
Brief Description	An overview of Deep Neural Networks, exploring architectures utilizing multiple hidden layers for complex problem-solving.
Learning/Take Away	Gained an understanding of how deeper network architectures are structured to model complex, non-linear relationships.

Session 2

Field	Details
Date	6/29/2026
Day wise Session (Morning and afternoon)	Session 2 (Forenoon, 11.30 AM - 1.00 PM)
Name of the Resource Person	Person A
Topic Covered	Regularization Techniques
Brief Description	A detailed discussion on various regularization methods designed to prevent overfitting in machine learning models.
Learning/Take Away	Learned how to apply standard techniques to improve model generalization and ensure robust performance on unseen data.

Session 3

Field	Details
Date	6/29/2026

Day wise Session (Morning and afternoon)	Session 3 (Afternoon, 2.00 PM - 3.30 PM)
Name of the Resource Person	Person A
Topic Covered	Building and training a simple DNN
Brief Description	A walkthrough of the foundational steps required to construct, compile, and initiate the training process for a simple Deep Neural Network.
Learning/Take Away	Acquired the knowledge necessary to set up the basic architecture and structure the training loop for a deep learning model.

Session 4

Field	Details
Date	6/29/2026
Day wise Session (Morning and afternoon)	Session 4 (Afternoon, 4.00 PM - 5.30 PM)
Name of the Resource	Person A

Person	
Topic Covered	Hands-on: DNN Implementation
Brief Description	A practical lab session focused on writing code and implementing a functional Deep Neural Network using standard programming libraries.
Learning/Take Away	Successfully developed and executed a working DNN script, translating theoretical network architecture concepts into practical code.

Day 4

Session 1

Field	Details
Date	6/30/2026
Day wise Session (Morning and afternoon)	Session 1 (Forenoon, 9.30 AM - 11.00 AM)
Name of the Resource Person	Person A
Topic Covered	Convolutional Neural Networks (CNNs): Concepts and Layers
Brief Description	An introduction to the architecture of CNNs, detailing the functionality of convolutional layers, pooling layers, and overall spatial data processing.
Learning/Take Away	Understood how CNNs efficiently extract features from images and grid-like data using specialized layer concepts.

Session 2

Field	Details

Date	6/30/2026
Day wise Session (Morning and afternoon)	Session 2 (Forenoon, 11.30 AM - 1.00 PM)
Name of the Resource Person	Person A
Topic Covered	Applications of CNNs in Computer Vision
Brief Description	An exploration of real-world use cases where Convolutional Neural Networks are heavily utilized within the computer vision domain.
Learning/Take Away	Gained insight into how CNNs power practical technologies such as object detection, facial recognition, and image segmentation.

Session 3

Field	Details
Date	6/30/2026
Day wise Session (Morning and afternoon)	Session 3 (Afternoon, 2.00 PM - 3.30 PM)

Name of the Resource Person	Person A
Topic Covered	Hands-on: Building a CNN for Image Classification
Brief Description	A practical, guided coding session focused on constructing and compiling a Convolutional Neural Network to classify a dataset of images.
Learning/Take Away	Successfully built and trained a functioning CNN model capable of categorizing visual data.

Session 4

Field	Details
Date	6/30/2026
Day wise Session (Morning and afternoon)	Session 4 (Afternoon, 4.00 PM - 5.30 PM)
Name of the Resource Person	Person A
Topic Covered	Case Study: Computer Vision

Brief Description	An in-depth analysis of a specific computer vision case study, highlighting the practical challenges and deployment strategies in the industry.
Learning/Take Away	Learned how theoretical computer vision models are adapted and deployed to solve complex, real-world industry problems.

Day 5

Session 1

Field	Details
Date	7/1/2026
Day wise Session (Morning and afternoon)	Session 1 (Forenoon, 9.30 AM - 11.00 AM)
Name of the Resource Person	Person A
Topic Covered	Introduction to RNNs: LSTM and GRU
Brief Description	An introductory session on Recurrent Neural Networks (RNNs), focusing on the specific architectures of Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU).
Learning/Take Away	Understood how RNN architectures handle sequential data and how LSTMs and GRUs mitigate issues like the vanishing gradient problem.

Session 2

Field	Details
Date	7/1/2026
Day wise Session (Morning and afternoon)	Session 2 (Forenoon, 11.30 AM - 1.00 PM)
Name of the Resource Person	Person A
Topic Covered	Sequence Modeling
Brief Description	A comprehensive look at sequence modeling techniques, focusing on processing data where order, context, and time dependencies are critical.
Learning/Take Away	Gained the foundational knowledge required to prepare and model time-series or context-dependent sequential data sets.

Session 3

Field	Details
Date	7/1/2026

Day wise Session (Morning and afternoon)	Session 3 (Afternoon, 2.00 PM - 3.30 PM)
Name of the Resource Person	Person A
Topic Covered	Hands-on: Text Models
Brief Description	A practical, guided coding session dedicated to constructing and implementing basic text-based models using sequential network structures.
Learning/Take Away	Successfully built and executed a text processing model, gaining practical experience in structuring sequential data workflows.

Session 4

Field	Details
Date	7/1/2026
Day wise Session (Morning and afternoon)	Session 4 (Afternoon, 4.00 PM - 5.30 PM)
Name of the Resource Person	Person A

Topic Covered	NLP with Deep Learning
Brief Description	An exploration of how deep learning architectures are applied to solve complex Natural Language Processing (NLP) tasks.
Learning/Take Away	Learned how modern deep learning approaches elevate the capabilities of NLP applications such as translation, sentiment analysis, and text generation.

Day 6

Session 1

Field	Details
Date	7/2/2026
Day wise Session (Morning and afternoon)	Session 1 (Forenoon, 9.30 AM - 11.00 AM)
Name of the Resource Person	Person A
Topic Covered	Introduction to Generative AI
Brief Description	An introductory overview of Generative AI concepts, focusing on how models can learn patterns to generate new, original content and data.
Learning/Take Away	Gained a foundational understanding of generative models and their primary distinction from traditional discriminative models.

Session 2

Field	Details

Date	7/2/2026
Day wise Session (Morning and afternoon)	Session 2 (Forenoon, 11.30 AM - 1.00 PM)
Name of the Resource Person	Person A
Topic Covered	GANs (Architecture & Training)
Brief Description	A detailed exploration of Generative Adversarial Networks (GANs), covering the internal architecture of generators and discriminators and the adversarial training loop.
Learning/Take Away	Understood the core mechanics of how adversarial training allows models to optimize and generate highly realistic synthetic data.

Session 3

Field	Details
Date	7/2/2026
Day wise Session (Morning and	Session 3 (Afternoon, 2.00 PM - 3.30 PM)

afternoon)	
Name of the Resource Person	Person A
Topic Covered	Building a basic GAN for Image Generation
Brief Description	A practical, hands-on session focused on the implementation and structural setup of a foundational Generative Adversarial Network designed for creating images.
Learning/Take Away	Successfully coded the components of a basic GAN, grasping how to link the generator and discriminator within a unified programming framework.

Session 4

Field	Details
Date	7/2/2026
Day wise Session (Morning and afternoon)	Session 4 (Afternoon, 4.00 PM - 5.30 PM)
Name of the Resource Person	Person A

Topic Covered	Image Generation
Brief Description	A continuation of practical applications, dedicated to executing the GAN model, generating visual outputs, and analyzing the results.
Learning/Take Away	Learned how to run a generative model for content creation and effectively evaluate the quality of the artificially generated images.

Day 7

Session 1

Field	Details
Date	7/3/2026
Day wise Session (Morning and afternoon)	Session 1 (Forenoon, 9.30 AM - 11.00 AM)
Name of the Resource Person	Person A
Topic Covered	Autoencoders & Variational Autoencoders
Brief Description	An overview of autoencoder architectures, covering standard autoencoders for dimensionality reduction and variational autoencoders for generative tasks.
Learning/Take Away	Understood how these networks learn efficient data encodings and map inputs to a probabilistic latent space for generating new samples.

Session 2

Field	Details

Date	7/3/2026
Day wise Session (Morning and afternoon)	Session 2 (Forenoon, 11.30 AM - 1.00 PM)
Name of the Resource Person	Person A
Topic Covered	Diffusion Models (Intro)
Brief Description	An introduction to diffusion models, detailing the forward mathematical process of adding noise and the reverse process used to generate new data.
Learning/Take Away	Gained a foundational understanding of the principles driving state-of-the-art diffusion models for high-quality image generation.

Session 3

Field	Details
Date	7/3/2026
Day wise Session (Morning and afternoon)	Session 3 (Afternoon, 2.00 PM - 3.30 PM)

Name of the Resource Person	Person A
Topic Covered	Hands-on: Autoencoder
Brief Description	A guided programming session focused on structuring, training, and validating a functional autoencoder using Python frameworks.
Learning/Take Away	Successfully built and trained an autoencoder model, translating the concepts of encoding and decoding layers into working code.

Session 4

Field	Details
Date	7/3/2026
Day wise Session (Morning and afternoon)	Session 4 (Afternoon, 4.00 PM - 5.30 PM)
Name of the Resource Person	Person A
Topic Covered	Hands-on: Image Denoising / Reconstruction

Brief Description	A practical application lab demonstrating how to utilize neural networks specifically for restoring corrupted data and removing noise from images.
Learning/Take Away	Learned how to practically apply deep learning techniques to solve real-world image restoration and denoising challenges.

Day 8

Session 1

Field	Details
Date	7/4/2026
Day wise Session (Morning and afternoon)	Session 1 (Forenoon, 9.30 AM - 11.00 AM)
Name of the Resource Person	Person A
Topic Covered	Transformers & Large Language Models
Brief Description	An overview of the Transformer architecture and its critical role in powering modern Large Language Models (LLMs).
Learning/Take Away	Gained an understanding of self-attention mechanisms and how foundational models process and generate human language.

Session 2

Field	Details

Date	7/4/2026
Day wise Session (Morning and afternoon)	Session 2 (Forenoon, 11.30 AM - 1.00 PM)
Name of the Resource Person	Person A
Topic Covered	Prompt Engineering
Brief Description	A comprehensive look at the techniques and methodologies for structuring inputs to effectively guide and constrain LLM outputs.
Learning/Take Away	Learned how to design and optimize prompts to elicit accurate, context-aware, and highly relevant responses from AI systems.

Session 3

Field	Details
Date	7/4/2026
Day wise Session (Morning and afternoon)	Session 3 (Afternoon, 2.00 PM - 3.30 PM)

Name of the Resource Person	Person A
Topic Covered	Hands-on: LLM APIs
Brief Description	A practical programming session exploring how to integrate and interact with large language models through commercial and open-source APIs.
Learning/Take Away	Successfully connected to an LLM interface programmatically, enabling automated data querying and text generation workflows.

Session 4

Field	Details
Date	7/4/2026
Day wise Session (Morning and afternoon)	Session 4 (Afternoon, 4.00 PM - 5.30 PM)
Name of the Resource Person	Person A
Topic Covered	Build AI Chatbot

Brief Description	A hands-on project dedicated to constructing a functional, interactive AI chatbot leveraging the APIs and models discussed throughout the day.
Learning/Take Away	Acquired the practical programming skills required to piece together an interactive conversational agent from scratch.